August 28, 2017 Matt Landreman

Title

This note is a suggestion from Neil Pomphrey.

I'm working in

~/Box Sync/work17/20170830-01-testing_VMEC_signs_using_a_simple_tokamak

First I make a mgrid file consisting of 2 coil groups: a TF coil (which can be a single infinite wire along the symmetry axis) and a Helmholtz pair.

A Helmholtz pair has coils at x = \pi R/2 where R is the radius.

Here are some coils files that do the job:

```
periods 1
begin filament

0.0 0.0 0.0 1.0
0.0 0.0 1.0 1.0 1 TF
10.0e+0 0.0  5.0e+0 1.0 2 Helmholtz
10.0e+0 0.0 -5.0e+0 1.0 2 Helmholtz
end
```

This file results in B_phi ~ 7.07e-8 Tesla.

Or, here is a case in which I use a single square TF coil rather than an infinite wire:

```
periods 1
begin filament

0.0    0.0 -1000.0 1.0
0.0    0.0  1000.0 1.0
1000.0 0.0  1000.0 1.0
1000.0 0.0 -1000.0 1.0
0.0    0.0 -1000.0 1.0 1 TF10.0e+0 0.0  5.0e+0 1.0 2 Helmholtz
10.0e+0 0.0 -5.0e+0 1.0 2 Helmholtz
end
```

This file results in B_phi ~ 2.00e-7 Tesla. In both cases, makegrid indicates that B_phi is ***positive*** for the TF coil group. This means that phi must increase as you move counterclockwise when viewed from above. Equivalently, (R,phi,Z) is right-handed.

From Ampere's Law, we expect B_phi to be

$$B_\phi = \frac{\mu_0 I}{2\pi R} = \frac{4\pi\times 10^{-7}}{2\pi R} \approx 2\times 10^{-7} T \tag{1}$$

since I'm considering the region $R \approx 1m$. So let's work with the square TF coil, since I'm confused about the current normalization for the infinite-straight-wire case.

The mgrid files have B_Z ***positive*** for the Helmholtz coil group. So the Helmholtz currents are flowing in the +phi direction, as I'd expect. The magnitude of B_Z from the mgrid file is 8.99e-8 T. The B field at the center of a Helmholtz pair is supposed to be

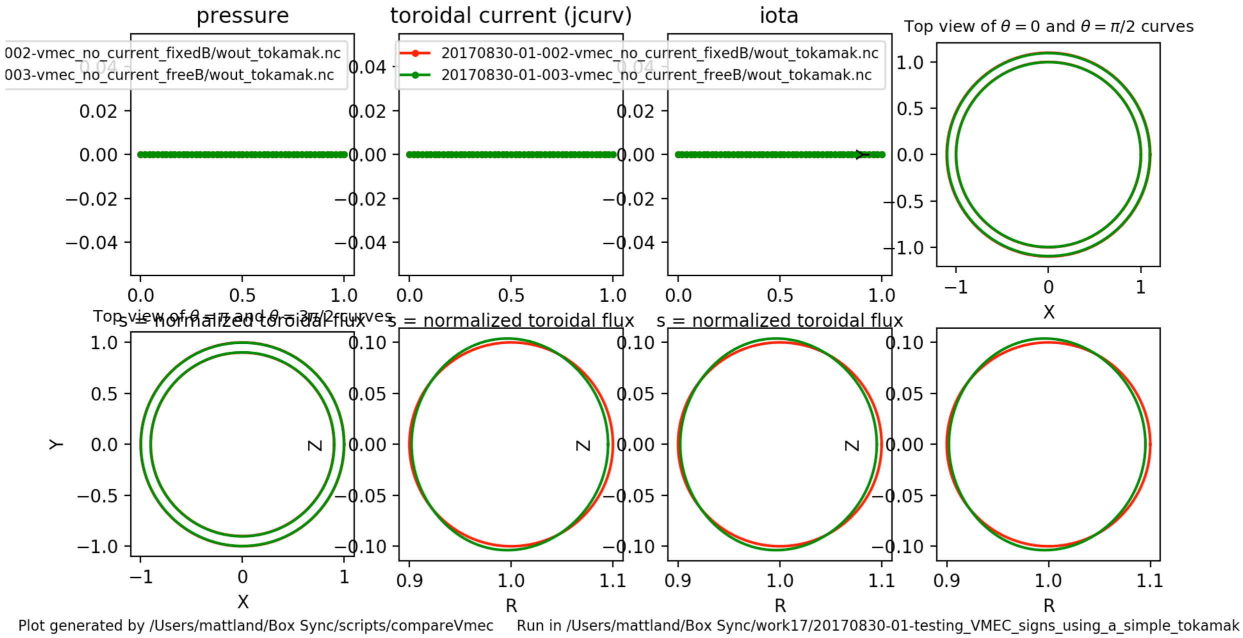
$$B = \frac{8}{5\sqrt{5}} \frac{\mu_0 I}{R} = \frac{8.99176\times 10^{-7} T}{(R/1m)} \tag{2}$$

I’m using R=10m for this Helmholtz pair, so B_Z in the mgrid file is precisely what I’d expect.

If the plasma minor radius is 0.1m, I expect the toroidal flux to be

$$B\pi r^2 = \left(2.0\times10^{-7}\right)\pi\left(0.1\right)^2 = \tag{3}$$

Comparing fixed vs free boundary, with no plasma current:



These runs used

```
  RBC(0,1) =   0.1      ZBS(0,1) =   0.1
```

So theta increases in the counterclockwise direction on the right cross-section of the tokamak. The Jacobian of (s,theta,phi) should be negative. B^v should be positive. I expect phi is positive and phips is negative. All of these check out with the results in the wout file.

In directory 004 and 005, I switched the direction in which theta increases:

```
RBC(0,1) =   0.1      ZBS(0,1) =   -0.1
```

The Jacobian of (s,theta,phi) should now be positive. However, when I ran VMEC, I found that it seems to have flipped the sign of rmnc(0.1), which is now ~ -0.1. So signgs is again negative. Weird.

Adding a toroidal plasma current

From eq (6.90) in Freidberg’s new MHD book, or (6.91) of Jeff’s original MHD book, the vertical field is

$$B_v = \frac{\mu_0 I}{4\pi R_0}\left(\beta_p + \frac{l_i - 3}{2} + \ln\frac{8R_0}{a}\right) \tag{4}$$

where the internal inductance is

$$l_i = \frac{2}{a^2 B_{\theta a}^2}\int_0^a dy\ yB_\theta^2\ . \tag{5}$$

Consider a uniform toroidal current density. Ampere's Law says

$$2\pi r B_\theta = \mu_0\left(\frac{r}{a}\right)^2 I_p \tag{6}$$

where I_n is the total current at the LCFS $r = a$. Then

$$B_\theta = \frac{\mu_0 I_p r}{2\pi a^2} \tag{7}$$

and

$$l_i = \frac{2}{a^4}\int_0^a dr\ r^3 = \frac{2}{a^4}\frac{a^4}{4} = \frac{1}{2}\ . \tag{8}$$

A uniform current density corresponds to

$$\frac{dI}{ds} = \frac{4\pi\left\langle j_{||}B\right\rangle}{c\left\langle B^2\right\rangle} \approx const \tag{9}$$

This makes sense because $s \propto r^2$ and $I \propto r^2$ so $I/s \propto r^0$.

Suppose we want $B_\theta / B \sim 0.1$ at the plasma edge. Then the plasma current we want is

$$0.1 = \frac{B_{\theta a}}{B} = \frac{\frac{\mu_0 I_p}{2\pi a}}{\frac{\mu_0 I_{TF}}{2\pi R}} = \frac{I_p}{I_{TF}}\frac{R}{a} = 10\frac{I_p}{I_{TF}} \tag{10}$$

So let's choose

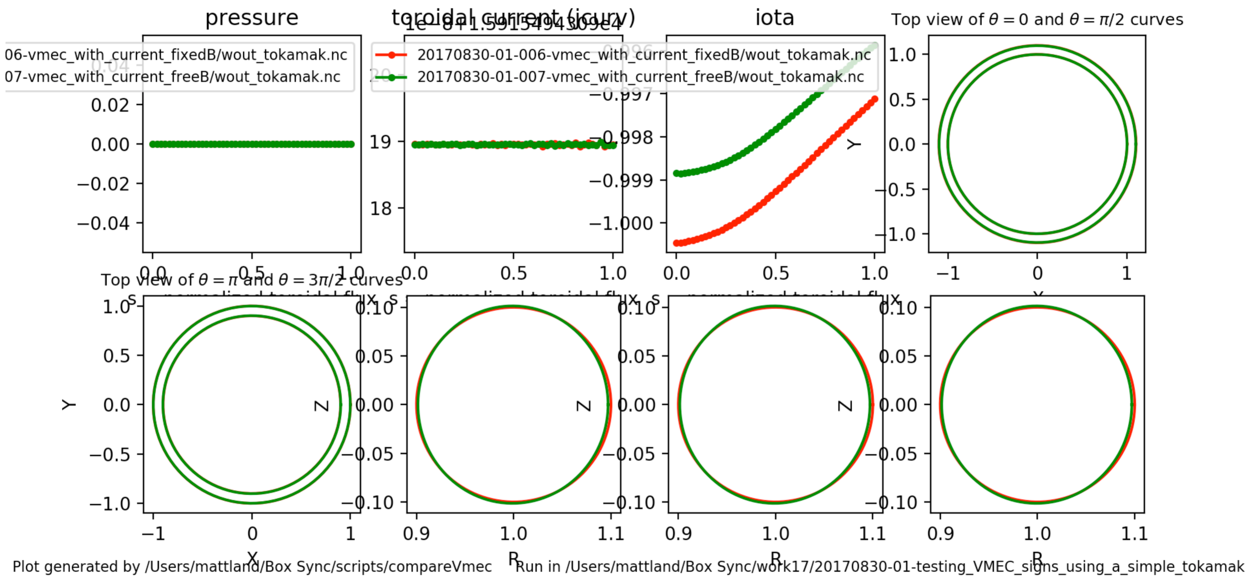
$$I_p = 0.01 \times I_{TF} \tag{11}$$

The necessary current in the Helmholtz coils is then given by equating (4) and (1):

$$\frac{\mu_0 I_p}{4\pi R_{plasma}}\left(0 + \frac{0.5-3}{2} + \ln 80\right) = \frac{8}{5\sqrt{5}}\frac{\mu_0 I_{Helmholtz}}{R_{Helmholtz}} \tag{12}$$

$$I_{Helmholtz} = I_p \frac{R_{Helmholtz}}{R_{plasma}}\frac{5\sqrt{5}}{32\pi}\left(0 + \frac{0.5-3}{2} + \ln 80\right) = \left(0.348322\right) I_p \frac{R_{Helmholtz}}{R_{plasma}} = \left(3.48322\right) I_p \tag{13}$$

For positive curtor, I had to set extcur(2) negative to get the free boundary calculation to work:



For run 007, signgs=-1. This is consistent with the fact that at the boundary, R = 1 + 0.1 cos(theta) and Z = 0.1 sin(theta). Phi is positive, as B_toroidal is in the grad zeta direction. Curtor is positive, which I think means that the current is in the grad zeta direction. If so, then B_theta should be in the negative grad theta direction, so bsupu and bsubu should both be negative. Indeed they are. Jdotb should be positive, and indeed it is. Bsupv and bsubv should both be positive, and they are.